



A novel laser micro/nano-machining system for FPD process

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ABSTRACT

Laser machining technology has potential to be adopted as micro- and nano-fabrication equipments in the field of flat panel display (FPD) industry. The equipments repair short, open or protrusion defects by cutting and welding using high-power laser. The equipments should be able to carry large sized mother glass and have high productivity and accuracy. To meet the requirements, the equipment should travel long range with higher speed and higher precision than the conventional. In this paper, a high precision decoupled dual-stage is proposed to transfer and position FPD mother glass. The dual-stage system consists of coarse stage actuated by linear motor and fine stage by moving magnet type voice coil motor. The control and design of the two stages are required to be considered independently if possible in order to take advantage of modular approach. In order to suppress disturbance from the coarse to fine stage, they are designed without mechanical connections. Dual-servo tracking controller is applied by adding fine controller to conventional coarse controller. Reaction force between fine and coarse stages is compensated by a force compensator (FC) because it is detrimental to positioning and scanning. By simulation and experiment, the performance of dual-stage is evaluated and compared.

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1. Introduction

Laser machining technology has potential to be applied to FPD industry. The equipments repair short, open or protrusion defects by cutting and welding by laser, which are found after display inspection in FPD process. In order to repair micro- and nano-pattern for FPD device such as plasma display panel (PDP) and liquid crystal display (LCD) as shown in Fig. 1, the equipments should be able to carry large sized mother glass and have high productivity and accuracy. To meet the requirements of next generation FPD process, the equipment should travel long range with higher speed and precision than the conventional.

Dual-servo XY stages are generally used to satisfy the requirements of equipments. The stage consists of coarse and

fine stages. The coarse stage moves the fine stage and laser head to long distance with high speed. The fine stage rejects the disturbance with wide bandwidth and high resolution. Linear motor is introduced as a coarse actuator and voice coil motor (VCM) as a fine actuator. Using the laser machining, the protrusion defect in color filter of LCD can be repaired. The size of defects is 10–30 μm in diameter and 5–10 μm in height.

The conventional dual-stages researched up to now has two stages connected in series, for example, a ball-screw drive or a linear motor (LM) and a PZT-driven flexure stage (Lee and Kim, 1997; Elfizy et al., 2005; Guo et al., 1998). In other words, the stages have been mechanically coupled with flexure structure. The performance of a coarse stage is generally limited by system vibration, thermal deformation, guide imperfection, mass, and frictions. These disturbances of a coarse stage

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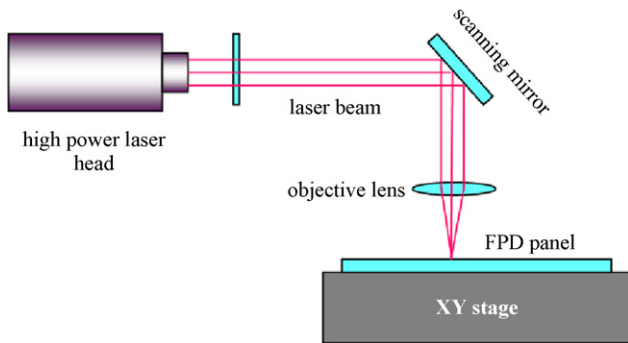


Fig. 1 – Working principle of a laser repair.

are directly delivered to a fine stage through the mechanical connections. In order to compensate these disturbances, the controller of fine stage should be complex.

These methods are not requiring the modular approach because design and control of two stages are inextricably interconnected inevitably. Especially, field engineers want to update their conventional single-servo LM stage toward dual-stage with appending fine stage. In the case, the modular approach of design and control is essential.

In this paper, we propose and design a new type of dual-stage: a mechanically decoupled-type dual-stage (MDDS). A linear motor (LM) and a moving magnet voice coil motor (MMVCM) are used as actuators for the stage of MDDS. The design and control of the two stages are carried out as independently as possible. Minimization of reaction force between the fine and coarse stages is achieved by no mechanical connections between the stages. Both stages are guided and suspended, respectively, by air bearings which float on a granite surface for frictionless movement.

By adding fine controller to conventional coarse controller, dual-servo tracking controller is realized for the modular approach philosophy. This paper applies a dual-servo algorithm. Despite the modular design, reaction force between fine and coarse stages is generated by magnetic coupling, which is detrimental to positioning and scanning even resulting in collision between the stages. Therefore, we propose a simple but effective force compensator (FC) that compensates the reaction force.

2. Design of mechanically decoupled dual-stage

2.1. Dual-stage configuration

As shown in Fig. 2, the proposed dual-stage is made up of a coarse stage which is actuated by LM and guided by air-bearing and a fine stage which is actuated by MMVCM and also guided by air-bearing. In general, LM has velocity ripples due to commutation mismatch between current and magnetic flux. VCM has smooth motions without velocity ripples because it does not need any commutations. It has longer working range than PZT actuator. Guide systems with guide bar, granite surface and air bearing are capable of sustaining a high load and make the dual-stage be frictionless. There is magnetic coupling but

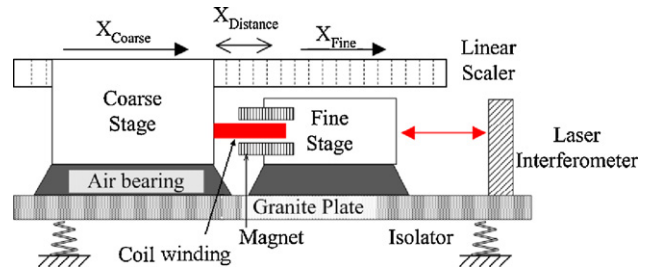


Fig. 2 – Concept of dual-stage.

no mechanical connection through the VCM because the coil windings are mounted on coarse stage and magnets attached to fine stage. The magnets are moving freely over the windings by Lorentz force. Due to the mechanically decoupled mechanism, the transferred disturbance from the coarse stage to the fine stage can be more minimized than the mechanism which has mechanical coupling. As position sensors of the coarse and fine stages, a linear scale and a laser interferometer are used, respectively.

In order to design, maintain, and update equipment, modular design approach is recommended. The system in Fig. 3 has two module, coarse stage with LM and fine stage with MMVCM. This modular approach is essential in workshop, to field engineer and for safety of the workpiece. The coarse and fine stages can work independently with their separate controller and position sensor. If one of them is broken, the other should work in workshop. The two stages can be fabricated and tested independently, which save lead time.

The LM of the coarse stage is an actuator for its hundreds of mm stroke. It consists of a stator (magnet) and a mover (coil winding). The stator is fixed on the guide bar of coarse stage. Therefore, x-motions of the coarse stage are decided by the displacements of the mover. And each VCM consists of coil windings which are attached to coarse stage and magnets on the fine stage. The fine stage is guided by air-bearing on granite surface, and actuated by four VCMs (x_1 , x_2 , y_1 , and y_2), respectively. Each VCM consists of a coil winding, four rare-earth magnets (Star Group, Ni coated N-40, $H_c = 923$ kA/m and $B_r = 1.25$ T), and two yokes made of permeable material, S10C.

Generally, moving coil type VCM are used in many industry fields: an optical disk drive, a hard disk drive, a linear motor, etc. Heat generation from the coil results in thermal deformation. On the contrary, a moving magnet type VCM

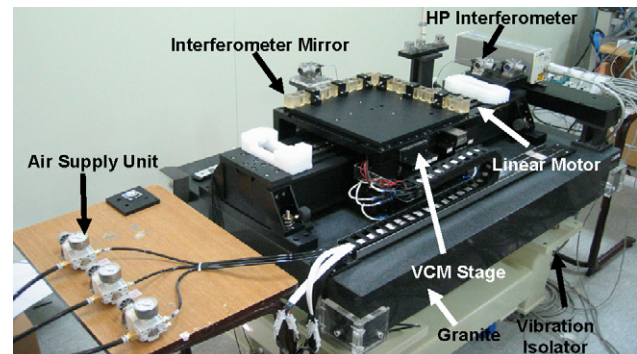


Fig. 3 – Fabricated mechanically decoupled dual-stage.

(MMVCM), whose coil winding is fixed, is used in MDDS. In this case, heat generation due to the coil windings is mainly dissipated through the air between magnet and coil winding. If the coil winding is mounted on the fine stage, the thermal disturbance transferred to the workpiece through the fine stage material, aluminium alloy because the thermal conductivity of the metal is higher than that of air about 10,000 times.

Due to the mechanical decoupling characteristic, the coarse and fine stages must follow the same trajectory and the relative distance between the two stages should be constantly maintained. When actuating the MDDS, reaction force between the fine and coarse stages is generated, which makes difficult maintain the distance and lets even collide between the two stages. Since workpieces such as PDP or LCD are placed on top of the fine stage for inspection motion, the performance of the decoupled-type dual-stage is mainly dependent on that of fine stage if the reaction compensated enough. The MDDS is fabricated and assembled as shown in Fig. 3.

2.2. Coarse stage modeling: linear motor

When an electric current is applied to coil in magnetic flux, Lorentz force generated on coil windings actuates the coarse stage guided by air-bearing. The coarse stage actuated by LM has electric and mechanical characteristics. If stiffness and damping are ignored and current driven method is chosen, the transfer function is simple as presented in Eq. (1) where m_{LM} is mass of mover and X_{LM} is the position of coarse stage. I is current and k_{fLM} is force constant of linear motor. Trilogy™ 410 coreless motor in Table 1 is used:

$$\frac{X_{LM}(s)}{I(s)} = \frac{k_{fLM}}{m_{LM}s^2} \quad (1)$$

2.3. Fine stage modeling: MMVCM

The fine stage is actuated by four VCMs. Two VCMs are for the x-direction and the other two VCMs for the y-direction. Specifications of VCM are described in Table 1.

The VCM, which is similar to linear motor, is governed by Eq. (2) due to Lorentz force. In the equation k_{fVCM} is force constant, I is current of MMVCM. Let “a” be the distance of two VCM actuating in the x-direction and “b” be that of two VCM actuating the y-direction. Then the equation of motion of the

Table 1 – Parameters of LM and VCM

Parameters	Value
LM	
m_{LM}	15 kg
k_{fLM}	54.5 N/A
K_E	61 Vs/m
L	10 mH
R	8 Ω
VCM	
m_{VCM}	26.5 kg
I_{VCM}	1.33 kg m ²
a	156 mm
b	375 mm
k_{fVCM}	27.22 N/A

fine stage is determined by Eq. (3) where m_{VCM} is the mass, I_{VCM} the moment of inertia, $F_{x1, x2}$ = x-axis force, and $F_{y1, y2}$ = y-axis force of the fine stage:

$$F = nBIl_{eff} = k_{fVCM}I \quad (2)$$

$$\begin{bmatrix} m_{VCM}\ddot{X}_{VCM} \\ m_{VCM}\ddot{Y}_{VCM} \\ I_{VCM}\ddot{\theta}_{VCM} \end{bmatrix} = \begin{bmatrix} \cos\theta & \cos\theta & -\sin\theta & -\sin\theta \\ \sin\theta & \sin\theta & \cos\theta & \cos\theta \\ \frac{b}{2} & \frac{-b}{2} & \frac{a}{2} & \frac{-a}{2} \end{bmatrix} \begin{bmatrix} F_{x1} \\ F_{x2} \\ F_{y1} \\ F_{y2} \end{bmatrix} \quad (3)$$

When current is applied to four VCMs, four forces are generated and the xy θ motions of the fine stage are controlled. The VCM system has a redundant actuator, but it is necessary for symmetry and safety. For simplicity, θ is assumed to be zero when the controller for VCM is designed.

3. Dual-servo controller

In order to introduce the modular approach strategy, the controllers of the coarse and fine actuators are separately proposed, designed, and embodied to the decoupled dual-stage system. PID controller is applied as a basic tracking controller of the stages.

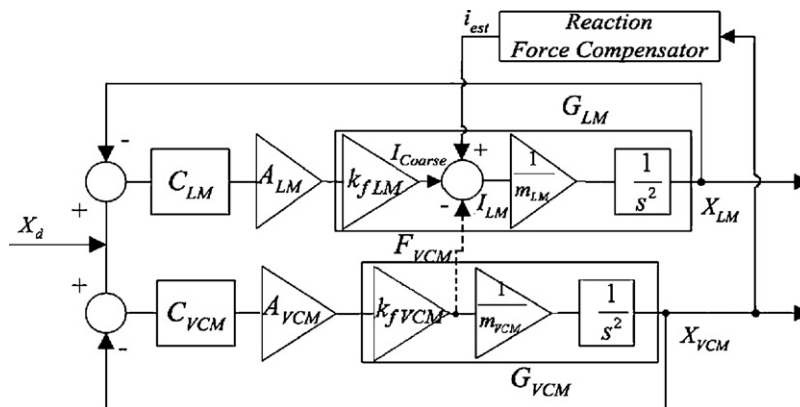


Fig. 4 – Block diagrams of dual-servo algorithms with FC.

3.1. Control algorithm for MDDS

In addition to the PID controller for each stage, a control algorithm for the MDDS is necessary for the proposed system. Especially, the coarse stage and fine stage must follow the same trajectory because the proposed dual-stage system is decoupled type with no mechanical coupling between two stages. From this point of view, a dual-stage algorithm is applied as shown in Fig. 4.

This algorithm is selected because it is general dual-servo control algorithm. The dynamic characteristics of the fine and coarse stages are similar, which enable coarse and fine stages to follow reference trajectory without collision even by the simple control algorithm. This is because the two motors of the stages have the same working principle. This is also good for modular design approach with independent design and control for the two stages, respectively.

3.2. Reaction force due to MMVCM

When the fine actuator is actuated, the reaction force with the same magnitude and opposite direction of VCM actuating force is also exerted on the coarse stage. However, because the coarse stage has no mechanical coupling with the fine stage and control damping only between the two stages, the reaction force induced by the fine to the coarse stage makes it difficult keep the relative distance between the two stages in wanted range. Fig. 4 shows VCM reaction forces are presented by dashed arrow.

F_{Reaction} can be estimated by the input current to VCMs or by the product of the mass and the acceleration of the fine stage. Since the acceleration could be easily calculated with use of the measured position of the fine stage, a force compensator (FC) is proposed to compensate the reaction force as shown in Fig. 4.

VCM reaction force to the coarse stage can be compensated by adding the same force to the coarse stage. F_{est} , VCM actuating force is estimated as Eq. (5). F_{est} is converted to current signal as shown in Eq. (6). After differentiating X_{VCM} with respect to time, the acceleration is filtered by low pass filter. In results, the compensated control input to coarse stage is represented in Eq. (7):

$$F_{\text{est}} = m_{\text{VCM}} \ddot{X}_{\text{VCM}} \quad (5)$$

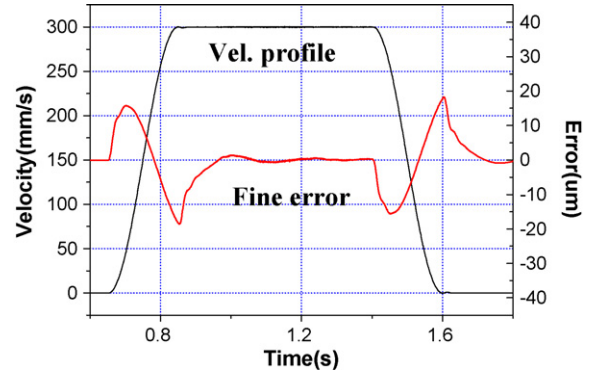


Fig. 5 – Velocity profile tracking error.

$$i_{\text{est}} = \frac{m_{\text{VCM}}}{k_{\text{fLM}}} \ddot{X}_{\text{VCM}} \quad (6)$$

$$I_{\text{LM}} = I_{\text{coarse}} + i_{\text{est}} \quad (7)$$

4. Experiments

In this section, the proposed dual-stage and control algorithms is evaluated for high speed and long stroke. And also the positioning accuracy of the MDDS is verified.

4.1. System configuration

The displacement of the coarse stage is measured by about 5 nm resolution in linear scale, Renishaw™ linear encoder and dSPACE™ 4096 interpolation board, and the displacement of the fine by a 5 nm laser interferometer of Hewlett Packard™ and dSPACE™ 4003 DIO board. These systems are controlled by dSPACE™ 1005 real time controller. The linear motor is actuated by a LA 400 linear servo current amplifier of Varedan™, and the fine VCM by a TA115 linear current amplifier of Trust Automation™.

4.2. Experimental results

The dual-servo control algorithm is embodied into the proposed dual-stage system. Firstly, we designed smooth reference velocity profile for smooth track following control as

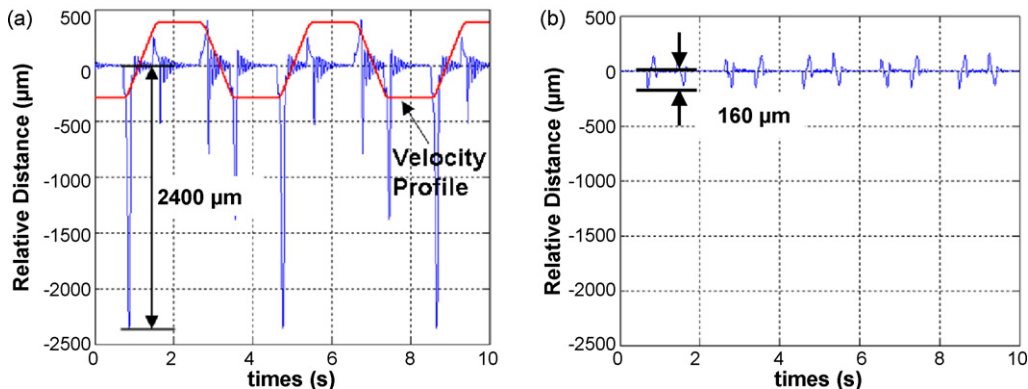


Fig. 6 – Relative distances between coarse and fine stage: (a) relative distance without FC and (b) relative distance with FC.

shown in Fig. 5. The trajectory is combination of sinusoidal curves so that it is continuous and has no jerk of acceleration.

Fig. 6 also shows the experimental results for the relative distance. Fig. 6(a) shows the result without FC. And results with FC are shown in Fig. 6(b). The relative distance is reduced from 2400 to 160 μm by using FC. The proposed algorithm is so effective that we can control the MDDS with high precision and high speed.

The velocity profile in Fig. 5 as their reference inputs has $0.25 \times g$ acceleration/deceleration region and 300 mm/s constant velocity region. This is the very high speed and acceleration. Tracking error is smaller than 30 μm . Resolution of the MDDS motion is about 10 nm. Displacement by a single velocity profile is about 375 mm, so the total displacement in Fig. 6(a) is about 1.2 m which is suitable for 5G LCD mother glass.

5. Conclusions

This paper presents and introduces the design and control for an ultra-precision dual-stage system for laser machining equipments applied to FPD process. The dual-stage is decoupled type which has no mechanical coupling between the coarse and fine stages. Minimization of reaction force between the fine and coarse stages is achieved by no mechanical connections between the stages. The controller and position sensor of the two stages are operated separately. This modular design approach makes it easy to developing a new dual-servo stage, to maintain a developed stage or to update a

conventional single servo stage. A dual-servo control algorithm has also been applied. The controller for the coarse and fine stage is modeled independently, which results in reaction force between fine and coarse stages. This is detrimental to positioning and scanning. Therefore, we propose force compensator to compensate reaction force applied to the coarse stage, and the effectiveness was demonstrated. The performance of proposed MDDS is high speed of 300 mm/s, high precision of 10 nm, long stroke of 1.2 m. These are suitable for the process of 5G LCD mother glass.

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